Abstract linear algebra homework9

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Problem 1

We verify the inner product axioms.

For all $f, g \in V$,

$$\langle f, g\rangle = \int_a^b f(t)g(t)\,dt = \int_a^b g(t)f(t)\,dt = \langle g, f\rangle.$$

For $f, h, g \in V$, by linearity of the integral,

$$\langle f + h,\, g\rangle = \int_a^b (f(t) + h(t))g(t)\,dt = \int_a^b f(t)g(t)\,dt + \int_a^b h(t)g(t)\,dt = \langle f, g\rangle + \langle h, g\rangle.$$

For $\lambda \in \mathbb{R}$,

$$\langle \lambda f, g\rangle = \int_a^b \lambda f(t)g(t)\,dt = \lambda \int_a^b f(t)g(t)\,dt = \lambda\langle f, g\rangle.$$

(Together with symmetry, this also gives additivity and homogeneity in the second slot, so $\langle\cdot,\cdot\rangle$ is bilinear.)

For any $f \in V$, $f(t)^2 \geq 0$ for all $t \in [a, b]$, so

$$\langle f, f\rangle = \int_a^b f(t)^2\,dt \geq 0,$$

since the integral of a nonnegative continuous function is nonnegative.

Clearly if $f \equiv 0$ then $\langle f, f\rangle = 0$. Conversely, suppose $f \in V$ and $f \not\equiv 0$; By calculas, since $f(t)^2$ is bigger than 0, the integral from a to b is also bigger than 0. Hence $\langle f, f\rangle > 0$ whenever $f \neq 0$.

So $\langle\cdot,\cdot\rangle$ is an inner product on V. ■

Problem 2

Let V be a real or complex inner product space, and let $v, w \in V$.

(a)

We first expand $\|v+w\|^2$ using the properties of the inner product:
$\|v+w\|^2 = \langle v+w,\, v+w\rangle = \langle v,v\rangle + \langle v,w\rangle + \langle w,v\rangle + \langle w,w\rangle = \|v\|^2 + \|w\|^2 + \langle v,w\rangle + \overline{\langle v,w\rangle}$,
using $\langle w,v\rangle = \overline{\langle v,w\rangle}$. Since $z+\bar{z} = 2\,\mathrm{Re}(z)$ for any complex number z,

$$\|v+w\|^2 = \|v\|^2 + \|w\|^2 + 2\,\mathrm{Re}\langle v,w\rangle.$$

Now $\mathrm{Re}(z) \le |z|$ for any complex number z, so

$$\mathrm{Re}\langle v,w\rangle \le |\langle v,w\rangle| \le \|v\|\,\|w\|,$$

where the last inequality is the Cauchy–Schwarz inequality. Substituting,

$$\|v+w\|^2 \le \|v\|^2 + \|w\|^2 + 2\|v\|\|w\| = (\|v\| + \|w\|)^2.$$

Since both sides of $\|v+w\|^2 \le (\|v\| + \|w\|)^2$ are nonnegative, we may take square roots to conclude

$$\|v+w\| \le \|v\| + \|w\|. \qquad \blacksquare$$

(b)

As above,

$$\|v+w\|^2 = \|v\|^2 + \|w\|^2 + \langle v,w\rangle + \langle w,v\rangle.$$

Similarly,

$$\|v-w\|^2 = \langle v-w, v-w\rangle = \|v\|^2 + \|w\|^2 - \langle v,w\rangle - \langle w,v\rangle.$$

Adding these two equations, the $\langle v,w\rangle$ and $\langle w,v\rangle$ terms cancel:

$$\|v+w\|^2 + \|v-w\|^2 = 2\|v\|^2 + 2\|w\|^2 = 2\left(\|v\|^2 + \|w\|^2\right). \qquad \blacksquare$$

Problem 3

Let V be a real inner product space and let $(e_1,\dots,e_n)$ be an orthonormal list in V such that every $v \in V$ satisfies

$$v = \sum_{i=1}^{n} \langle v, e_i\rangle e_i. \tag{$*$}$$

We show $(e_1,\dots,e_n)$ is a basis for V.

Spans V. Equation $(*)$ says every $v \in V$ is a linear combination of $e_1,\dots,e_n$ (with coefficients $\langle v,e_i\rangle$). Hence $\mathrm{span}(e_1,\dots,e_n) = V$.

Linearly independent. Suppose $a_1,\dots,a_n \in \mathbb{R}$ satisfy

$$a_1e_1 + \cdots + a_ne_n = 0.$$

Fix $j \in \{1,\dots,n\}$ and take the inner product of both sides with e_j:

$$0 = \langle 0, e_j\rangle = \Big\langle \sum_{i=1}^{n} a_ie_i,\, e_j \Big\rangle = \sum_{i=1}^{n} a_i\langle e_i,e_j\rangle = a_j,$$

since $\langle e_i,e_j\rangle = 0$ for $i \ne j$ and $\langle e_j,e_j\rangle = 1$ by orthonormality. As j was arbitrary, $a_1 = \cdots = a_n = 0$. Hence $(e_1,\dots,e_n)$ is linearly independent.

Since $(e_1,\dots,e_n)$ spans V and is linearly independent, it is a basis of V. $\blacksquare$

Check-In

Hello Akash! I am summer visiting and also I will be in year two this fall. Since I will not be in Berkeley this year, it is not neccesary to ask what course I want to take in Berkeley in this fall. But in the sense of math, I still want to figure out the things or structure I should learn. So for me in the future, I really want to get into the field of algebraic geometry/topology. However , a lot of teacher told me(include professors, phd and my high school teacher) that I don't need to be to early to determine the thing that I focus on. So I really want to know your suggestion.

Also, for my own studying plan. I have learned abstract algebra in my first year, but I don't think I make it good, so I audit MAT113 here to make sure the knowledge. I currently working on category theory and just started Limits and colimits. And also commutative algebra which I use Atiya's book but i just get to the modules part. Next semster I want to do some independent study but I still not reach to my academic advisor so before that I want to ask if you have any suggestions.

For further consideration, I might exchange to Berkeley in my third year for one semster. Some one suggest me the Borcherds's course of algebra which is really good. So I also want to know your suggestion if I exchange to Berkeley in my third year, which semester should I, and also if you have any recommandation on the Berkeley regular semester's course/ graduate course.

Sorry for writing to much. Also I know it is very very difficult to apply Berkeley pure mathematics phd, especially algerba stuffs, I want to know if you have any suggestions on my study plan and any "technics" or other words what should I reach or what should I must do if I want to apply to Berkeley phd.

Last but not least, it is a very nice time to learn from you. Wish you a happy life and your academically successful!!!